

Assembly Instructions

Gas engine-generator sets

**MTU 8V4000 GS – GG08V4000A1/A2 with engines 8V4000L32/L32FB/L33
and 8V4000L64/L64FNER**

**MTU 12V4000 GS – GG12V4000A1/A2 with engines 12V4000L32/L32FB/L33
and 12V4000L64/L64FNER**

**MTU 16V4000 GS – GG16V4000A1/A2 with engines 16V4000L32/L32FB/L33
and 16V4000L64/L64FNER**

**MTU 20V4000 GS – GG20V4000A1/A2 with engines 20V4000L32/L32FB/L33
and 20V4000L64/L64FNER**

M060808/03E

Name	Model number	Engine model	Application group
MTU 8V4000 GS	GG08V4000A1	8V4000L32 8V4000L32FB 8V4000L33 8V4000L64 8V4000L64FNER	3A, continuous operation, unrestricted
	GG08V4000A2	8V4000L32 8V4000L32FB 8V4000L33 8V4000L64 8V4000L64FNER	3A, continuous operation, unrestricted
MTU 12V4000 GS	GG12V4000A1	12V4000L32 12V4000L32FB 12V4000L33 12V4000L64 12V4000L64FNER	3A, continuous operation, unrestricted
	GG12V4000A2	12V4000L32 12V4000L32FB 12V4000L33 12V4000L64 12V4000L64FNER	3A, continuous operation, unrestricted
MTU 16V4000 GS	GG16V4000A1	16V4000L32 16V4000L32FB 16V4000L33 16V4000L64 16V4000L64FNER	3A, continuous operation, unrestricted
	GG16V4000A2	16V4000L32 16V4000L32FB 16V4000L33 16V4000L64 16V4000L64FNER	3A, continuous operation, unrestricted
MTU 20V4000 GS	GG20V4000A1	20V4000L32 20V4000L32FB 20V4000L33 20V4000L64 20V4000L64FNER	3A, continuous operation, unrestricted
	GG20V4000A2	20V4000L32 20V4000L32FB 20V4000L33 20V4000L64 20V4000L64FNER	3A, continuous operation, unrestricted

Table 1: Applicability

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Preface

This installation manual is intended to assist project planners, system/component designers and installation personnel in planning or carrying out the installation of MTU components.

The installation manual is applicable to all gas engines from the 4000 series (see business portal).

The objective of the installation manual is to facilitate the assembly of components.

The installation manual does not relieve those responsible for the system from independently ensuring that installation is carried out in an appropriate manner.

In addition to the assembly instructions, the following documents included in the delivery must be observed:

- Planning drawing
- Schematic diagrams
- TEN data
- Arrangement drawings
- Operating manual (see business portal)
- Maintenance schedule (see business portal)
- Fluids and lubricants specifications (see business portal)
- MTU standards

Each document is taken from the respective specifications document required for the order.

Adherence to the prescribed schedule for maintenance tasks will have a positive effect on operational safety, reliability and long service life. During the planning or installation of the plant system, ease of access for operation, maintenance and repair work is to be ensured.

Table of Contents

1	Safety		4	Installation Conditions	
1.1	Important provisions for all products	7	4.1	Installation room - Requirements	60
1.2	MTU Onsite Energy GmbH gas systems - Intended use	9	4.2	Assembly - Preparatory work	61
1.3	Personnel and organizational requirements	11	4.3	Engine-generator set - Lifting and installation	63
1.4	Safety regulations for initial start-up and operation	12	4.4	Engine-generator set with split base frame - Lifting and installation	64
1.5	Safety regulations for commissioning and operation, specific information for gaseous fuel supply or and dual fuel applications	14	4.5	Optional Components	65
1.6	Safety regulations for assembly, maintenance, and repair work	16	4.5.1	Additional component provision	65
1.7	Safety regulations for assembly, maintenance and servicing work, specific information for gaseous fuel and dual fuel supply or applications	20	4.5.2	Heat module - Lifting and installation	66
1.8	Fire and environmental protection, fluids and lubricants	21	4.5.3	Cooler - Lifting and installation	67
1.9	Fire prevention and environmental protection, fluids and lubricants, auxiliary materials, specific information for gaseous fuel supply in gaseous-fuel or dual-fuel applications	24	4.5.4	Plate-core heat exchanger - Lifting and installation	68
1.10	Standards for warning notices in the text and highlighted information	25	4.5.5	Control cabinet - Installation	69
1.11	Conventions for warning notices on safety signs on the engine-generator set	26	5	Interface Description	
1.12	Safety signs on engine-generator set - Overview	28	5.1	Engine generator set - Interfaces	70
1.13	Overview of protective devices on the engine-generator set	30	5.2	Heat module - Interfaces	72
2	Transport		5.3	Control	74
2.1	Transport	32	6	Depreservation	
2.2	Transport locking device for engine mounting	39	6.1	Depreservation	75
2.3	Transport locking device of generator mounting	40	7	Mechanical Installation	
2.4	Crankshaft transport locking device	42	7.1	Cooling System and Heating Water System	76
2.5	Tie-down instructions	44	7.1.1	Coolant hoses - Installation requirements	76
3	Engine-Generator Set		7.1.2	Engine coolant system - Connection	79
3.1	Engine-generator set - Nameplate and designation	46	7.1.3	Mixture cooling system - Connection	81
3.2	Engine-generator set - Standard scope of supply	49	7.1.4	Heat module - Connection	84
3.3	Engine-generator set - Optional scope of supply	59	7.1.5	Heating water circuit - Connection	86
			7.1.6	Cooler - Connection	87
			7.1.7	Exhaust gas heat exchanger - Connection	89
			7.1.8	Emergency cooler - Connection	90
			7.1.9	Coolant-conveying lines - Check	91
			7.1.10	Insulation for cooling system - Installation	93
			7.2	Gas System	94
			7.2.1	Gas train - Prerequisites for installation	94
			7.2.2	Gas train - Installation	96
			7.2.3	Gas train - Connection	99
			7.2.4	Gas system - Check	100
			7.3	Lube Oil System	102
			7.3.1	Oil flexible metal tubes - Prerequisites for installation	102
			7.3.2	Lube oil system - Connection	103
			7.3.3	Oil-conveying lines - Check	105
			7.4	Exhaust System	107
			7.4.1	Exhaust system - Prerequisites for installation	107

7.4.2 Exhaust pipe bellows - Installation	108	8.9 Intrinsically safe circuits - Connection	154
7.4.3 Exhaust silencer - Connection	110	8.10 Equipotential bonding and protective conductor (grounding) - Connection	155
7.4.4 Catalytic converter for 4000 Series - Installation	113	8.11 Cabling - Check	157
7.4.5 Exhaust gas heat exchanger - Installation	115	8.12 Cable list	158
7.4.6 Exhaust lines - Check	118	8.13 Correct cabling	161
7.4.7 Insulation for exhaust system - Installation	119	8.14 Electromagnetic compatibility (EMC)	165
7.5 Engine-Generator Set Room	121	9 Supplementary Technical Information	
7.5.1 Room temperature sensor - Connection	121	9.1 Lines - Flushing	167
7.5.2 Gas alarm system - Connection	122	9.2 Initial operation - Prerequisites	168
7.5.3 Fire detection system - Connection	124		
7.5.4 Flashing horn - Connection	126		
8 Electrical Installation		10 Appendix A	
8.1 Wiring - Requirements	127	10.1 Abbreviations	169
8.2 Power cable - Installation	129	10.2 Standards and directives to be complied with	170
8.3 Power cables on generator - Connection	134	10.3 MTU Onsite Energy contact person / service partner	172
8.4 Generator circuit breaker (control part) - Connection	137		
8.5 MMC control cabinet - Connection	139	11 Appendix B	
8.6 X2X cable - Connection	140	11.1 Index	173
8.7 Starter battery - Connection	142		
8.8 Starter - Connection	152		

1 Safety

1.1 Important provisions for all products

General

This product may pose a risk of injury or damage in the following cases:

- Incorrect use
- Operation, maintenance and repair by unqualified personnel
- Changes or modifications which are neither made nor authorized by the manufacturer
- Noncompliance with the safety instructions and warning notices

Nameplates

The product is identified by nameplate, model designation or serial number. This data must match the specifications in these instructions.

Nameplates, model designation or serial number can be found on the product.

All EU-certified engines delivered by MTU come with a second nameplate. This second nameplate is delivered "loosely" with the engine. If the nameplate secured to the engine after installation in the vehicle/system is not visible without the removal of components, the system integrator must install the second nameplate in a clearly visible area on the vehicle/system.

Emission specifications and emission label

Responsibility for compliance with emission regulations

Modification or removal of any mechanical/electronic components or the installation of additional components including the execution of calibration processes that might affect the emission characteristics of the product are prohibited by emission regulations. Emission-related components must only be serviced, exchanged or repaired if the components used for this purpose are approved by the manufacturer.

Noncompliance with these specifications will invalidate the design type approval or certification issued by the emissions regulation authorities. The manufacturer does not accept any liability for violations of the emission regulations.

The product must be operated over its entire life cycle according to the conditions defined as "Intended use" (→ Page 9).

Emission certification applicable to engines with EPA Nonroad Tier 4 emission certification in accordance with 40 CFR 1039 and with EPA Locomotive emission certification in accordance with 40 CFR 1033

Extract from the standard:

Failing to follow these instructions when installing a certified engine in a piece of nonroad equipment violates federal law (40 CFR 1068.105(b)), subject to fines or other penalties as described in the Clean Air Act.

Extract from the standard:

If you install the engine in a way that makes the engine's emission control information label hard to read during normal engine maintenance, you must place a duplicate label on the equipment, as described in 40 CFR 1068.105.

When fitting the second label, the requirements of 40 CFR 1068.105(c) must be followed and observed. This paragraph describes the process for requesting and fitting the label, the documentation obligations and storage obligations for the required documents.

Replacing components with emission labels

On all MTU engines fitted with emission labels, these labels must remain on the engine throughout its operational life.

Exception: Engines used exclusively in land-based, military applications other than by US government agencies.

Please note the following when replacing components with emission labels:

- The relevant emission labels must be affixed to the spare part.
- Emission labels shall not be transferred from the replaced part to the spare part.
- The emission labels must be removed from the replaced part and destroyed.

1.2 MTU Onsite Energy GmbH gas systems – Intended use

Intended use

The machine is intended to be used exclusively for electrical power generation and waste heat utilization in stationary operation, either in sufficiently ventilated rooms or in container solutions.

A gas/air mixture is fed to the internal combustion reciprocating piston engine. This is converted into mechanical work in the individual cylinders through combustion. The flywheel of the engine drives the synchronous generator connected via an elastic coupling, resulting in a conversion to electric energy. The engine and the generator are joined to the base frame with elastic, vibration-damped elements. For mains frequency 60 Hz, depending on the series either a gearbox is used between the engine and generator (Series 4000) or the engine speed is increased accordingly (Series 400). The waste heat generated during the combustion process is dissipated to the surroundings via coolers (power generator) and/or made available to the customer via heat exchangers (combined heat and power plant).

Area of application	Suitable	Comment
Industry	Yes	–
Business/Service	Yes	In some individual cases, however, such as with swimming pool facilities, schools, heating stations for apartment buildings and the like, there is no end user in the classical sense involved.
Household	No	Operation is excluded
Potentially explosive area	No	Operation is not permitted in potentially explosive areas
Underground	No	Operation is excluded
Mobile use	No	Operation is excluded

Table 2: Area of application of machine

There are 3 operating modes of the machine:

- Grid parallel operation
- Mains backup mode
- Isolated operation

Foreseeable misuse

Any other use, particularly misuse, is considered as being contrary to the intended purpose. Such improper use increases the risk of injury and damage when working with the product. The manufacturer shall not be held liable for any damage resulting from improper use.

Operation

Operation is not permissible if the physical conditions specified on the technical data sheet or on/in other documents (concerning temperature, pressure, site altitude, etc.) are not complied with. Operation is not permissible without protective equipment, usage of approved spare parts, exclusion of conversions, warning of foreseeable misuse.

Operation is not permissible without the use of fluids and lubricants from MTU Onsite Energy (natural gas, biogas, quality-compliant gas, quality-compliant water, lubricants) in accordance with the (→ fluids and lubricants specifications).

Operation is not permissible without adopting the preservation measures approved by MTU Onsite Energy according to the (→ Preservation and Represervation Specifications).

The specifications of the manufacturer will be amended or supplemented as necessary. Prior to operation, make sure that the latest version of these are adopted. The latest version can be found at:

- <http://www.mtuonsiteenergy.com>

Surroundings, climate, temperature

The weather plays no role with regard to the intended use of the genset/module provided that the operation takes place inside an installation room. For container applications, however, the permissible outdoor temperatures at the installation site plays a decisive role. The product-specific ambient conditions such as climate and temperature are indicated in the technical description.

Prerequisite for proper use

Fulfillment of all the requirements set out in the “Installation conditions” document.

Any use that does not fall within the described limits for use is regarded as improper.

Modifications or conversions

Unauthorized changes to the product breach the conditions concerning its intended use and compromise safety.

Only modifications or conversions expressly authorized by the manufacturer are regarded as compliant in this sense. The manufacturer shall not be held liable for any damage resulting from unauthorized modifications or conversions.